

Resume–Job Matching & Talent-Marketplace Engine

Project Presentation

An intelligent platform for candidate–job matching, skill normalization and talent allocation

- Abstract
- Introduction
- Literature Survey
- Problem Statement and Objectives
- Hardware and Software Requirements
- Results
- Conclusion
- Future Scope

- The proposed engine automates matching between candidate résumés and job postings using structured skill and experience data.
- It normalizes different skill names and synonyms into a common vocabulary for more reliable comparison.
- A compatibility score ranks candidates against each role based on skills, experience, education and job requirements.
- An optimization module can assign suitable candidates to multiple roles while maximizing overall team fit.
- The platform also identifies the minimum combined skill set needed to staff a team, supporting faster and more informed hiring decisions.

Introduction

Hiring Challenge

Recruiters handle large volumes of résumés and job descriptions with inconsistent terminology and formats.

Proposed Solution

An intelligent engine extracts, normalizes and compares candidate skills with job requirements.

Marketplace Value

Candidates discover relevant opportunities while employers find better-fit talent faster.

- Reduces manual screening effort and improves consistency.
 - Provides transparent matching scores and ranked recommendations.
 - **Sorting algorithms** (e.g., quicksort/mergesort or heap-based top-K selection) — to rank candidates by compatibility score.
 - Supports both individual job matching and multi-role team staffing.
 - **Hashing / Hash Maps** — to map different skill names/synonyms (e.g., "JS", "Javascript") to one standard skill
- String similarity algorithm, Bipartite Graphs

Literature Survey

Keyword / Rule Matching

Traditional systems rely on exact keywords. They are simple but may miss equivalent skills and contextual relevance.

Semantic Matching

NLP and embedding-based approaches capture related terms and meaning beyond exact keyword overlap.

Optimization-Based Hiring

Assignment and set-cover techniques can allocate candidates to roles and minimize the skills required for a team.

- Gap identified: many solutions focus on ranking candidates but do not jointly address normalization, optimal assignment and team skill coverage.
- The proposed engine combines these capabilities into a unified talent-marketplace workflow.

Drawbacks vs Existing Systems (short):

- 1. No proven semantic AI – existing platforms (LinkedIn, Indeed) already use advanced NLP; yours only plans to add it.**
- 2. No real data/accuracy testing – no accuracy/precision numbers shown, just a workflow concept.**
- 3. Limited scalability – built for small-scale (4–8GB RAM), not millions of users like real platforms.**
- 4. Expensive algorithms – Set Cover & Assignment optimization are computationally heavy for large data.**
- 5. No bias/fairness checks yet – fairness is only planned for the future, not built now.**
- 6. Cold start issue – new users/jobs with little data may get poor matches.**
- 7. Static skill database – won't auto-update with new/niche skills like big platforms do.**
- 8. No explainability yet – can't currently tell *why* a candidate matched (planned for future)**

Problem Statement and Objectives

Problem Statement

Hiring platforms must efficiently match large candidate pools to roles despite inconsistent skill names, varied résumé formats and multiple staffing constraints.

Core Goal

Build an engine that scores candidate–job fit, recommends suitable talent, optimizes role assignment and reports required team skills.

- Extract and structure relevant résumé and job-posting information.
- Normalize skill vocabularies, synonyms and related terms.
- Compute a meaningful candidate-to-job compatibility score.
- Optimize assignment of candidates to multiple roles.
- Identify the minimum skill set required to staff a team.

Hardware and Software Requirements

Hardware Requirements

- Laptop / desktop computer
- Minimum 4 GB RAM (8 GB recommended)
- Dual-core processor or better
- Stable internet connection

Software Requirements

- Python / JavaScript development environment
- React for web interface
- Backend API framework
- Database for candidate, job and skill data

Processing & AI Layer

- NLP / text preprocessing
- Skill extraction and normalization
- Similarity or embedding model
- Ranking and optimization algorithms

Development Tools

- VS Code or equivalent IDE
- Git and GitHub
- Modern web browser
- API testing tool such as Postman

Results



- Produces ranked candidate recommendations instead of manual résumé-by-résumé screening.
- Makes skill equivalence easier to handle through normalization.
- Provides a foundation for measurable matching accuracy, staffing efficiency and reduced hiring effort.

Future Scope

Advanced AI Matching

Use transformer-based embeddings and LLMs for deeper semantic understanding of résumés and job descriptions.

Personalized Recommendations

Recommend jobs, candidates and learning paths based on skills, experience and career goals.

Real-Time Marketplace

Support live job updates, candidate availability, recruiter alerts and dynamic ranking.

Fairness & Explainability

Add bias monitoring and clear explanations for why a candidate matches a role.

Skill Gap Analysis

Identify missing skills and recommend targeted courses or certifications.

Enterprise Integration

Integrate with ATS, HR systems, interview scheduling and workforce analytics.

Conclusion

- The Resume–Job Matching & Talent-Marketplace Engine provides an end-to-end approach to intelligent talent discovery.
- Combining skill normalization, compatibility scoring and optimization improves the relevance of candidate recommendations.
- The system can reduce repetitive recruiter effort and help organizations make faster, data-driven staffing decisions.
- Its modular design allows future integration of advanced NLP, analytics and enterprise hiring workflows.